Design and Simulation of a 100 MHz High-Linearity Common-Source Amplifier

Diyush S
Department of Electrical Engineering
Indian Institute of Technology Kanpur
diyushs24@iitk.ac.in

Abstract

This report details the design evolution, architectural trade-offs, and final implementation of a high-linearity 100 MHz single-stage amplifier using a 180nm CMOS process node. The design targets strict harmonic constraints ($HD1/HD3 \geq 40$ dB) for a 1V peak-to-peak output swing across an AC-coupled 1 kΩ load under a 1.8V supply. The project followed an iterative development cycle: moving from a standard resistive-load Common-Source (CS) amplifier, through a CMOS active-load pair topology, and finally converging on an optimized single-transistor NMOS CS architecture with robust passive source degeneration. The ultimate design meets all physical boundaries and harmonic metrics, verified via the Cadence Virtuoso Analog Design Environment (ADE XL).

1 Introduction and Specifications

High-frequency analog integrated front-ends require robust amplification blocks capable of driving modest resistive loads while maintaining high linearity. In typical receiver architectures, harmonic distortion introduces unwanted spurious signals that degrade signal integrity. This design project systematically explores the topology space to satisfy the stringent requirements defined in the EE210 specification guidelines.

The design boundaries are defined as follows:

- **Load Resistance (R_L):** 1 kΩ, with one terminal grounded and the other AC-coupled to the amplifier stage.
- **Linearity Metric:** Harmonic suppression ratio $HD1/HD3 \geq 40$ dB for an output peak-to-peak voltage swing ($V_{o,\text{pk-pk}}$) of approximately 1V.
- **Center Frequency:** 100 MHz sinusoidal input excitation, centered around a 0V DC baseline.
- **Supply Voltage (V_{DD}):** Single 1.8V DC rail.
- **Component Constraints:** Passive inductors are strictly prohibited; individual capacitance limits are capped at 100 pF.

2 Design Evolution

Achieving simultaneous optimization of gain, output voltage headroom, and harmonic suppression required a multi-phase design progression.

2.1 Initial NMOS Common Source

The initial topology comprised a classic NMOS Common-Source amplifier with a passive drain resistor (R_D). While this choice offered predictable biasing behavior, it presented severe gain-linearity trade-offs. To deliver a 1V peak-to-peak output envelope across the load, the active transistor had to be driven with wide gate voltage swings. The non-linear transconductance (g_m) of the un-degenerated transistor generated significant third-order harmonic components ($HD3$), causing the distortion metric to fall far short of the mandatory 40 dB threshold.

2.2 Exploration of NMOS with PMOS Active Load

To break the performance bottlenecks of Phase 1, an alternative active-load architecture was developed.

Architecture Setup: A PMOS device (`pmos3`, $W = 200\ \mu$m, $L = 180$ nm) was introduced as an active current-source load for the primary NMOS transistor (`nmos3`, $W = 100\ \mu$m, $L = 180$ nm). A small source degeneration resistor ($R_4 = 10\ \Omega$) was placed beneath the NMOS source terminal. A complex passive network (R_1, R_0, R_2, R_3) was sized between 100 kΩ and 125 kΩ to control the gates of both active components simultaneously.

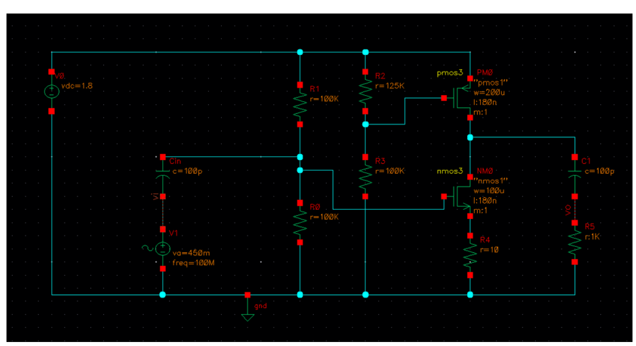

Figure 1: Schematic diagram of the NMOS with PMOS Active Load framework

Performance and Biasing Challenges: Theoretically, the high incremental output resistance ($r_{o,n} \parallel r_{o,p}$) of the active load provides substantial open-loop voltage gain and relaxes DC drop constraints, maximizing voltage headroom.

However, this topology introduced a severe biasing “Tug-of-War.” Due to the high sensitivity of the PMOS current source, minor variations in the drain operating point caused notable shifts in the transistor operating regions. When the bias network was modified to resolve output voltage clipping during a $V_a = 450$ mV input drive, the resulting operating point migration degraded the harmonic profile.

As verified in the transient and DFT simulation plots:

- **Fundamental Tone** **(**$HD1$**) at 100 MHz:** 572.67 mV
- **Third Harmonic** **(**$HD3$**) at 300 MHz:** 28.96 mV
- **Voltage Gain** **(**A_v**):** Given an input peak amplitude of $V_a = 450$ mV and a fundamental output amplitude of 572.67 mV, the large-signal voltage gain at the fundamental frequency is:

$$A_{v,\text{failed}} = \frac{572.669 \text{ mV}}{450 \text{ mV}} \approx 1.27 \text{ V/V } (2.09 \text{ dB}) \tag{1}$$

Evaluating the linearity performance in decibels yields:

$$\left(\frac{HD1}{HD3}\right)_{\text{failed}} = 20\log_{10}\left(\frac{572.669 \text{ mV}}{28.9623 \text{ mV}}\right) \approx 25.92 \text{ dB} \tag{2}$$

This performance proved insufficient, showing that the active load architecture could not maintain symmetry and linearity under high-swing conditions.

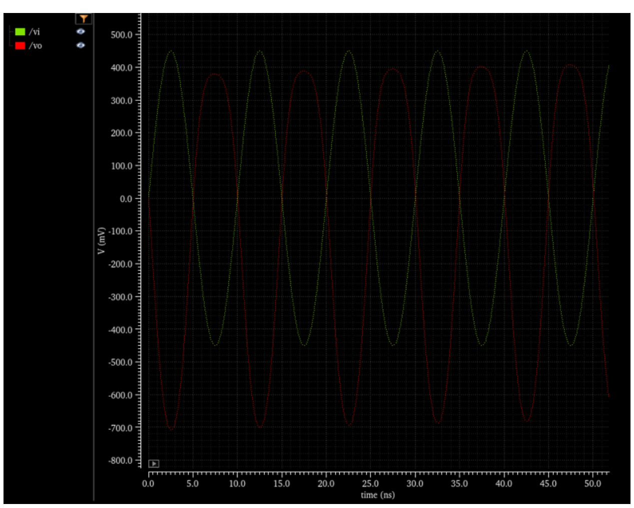

Figure 2: Transient waveform profile for the Active Load configuration

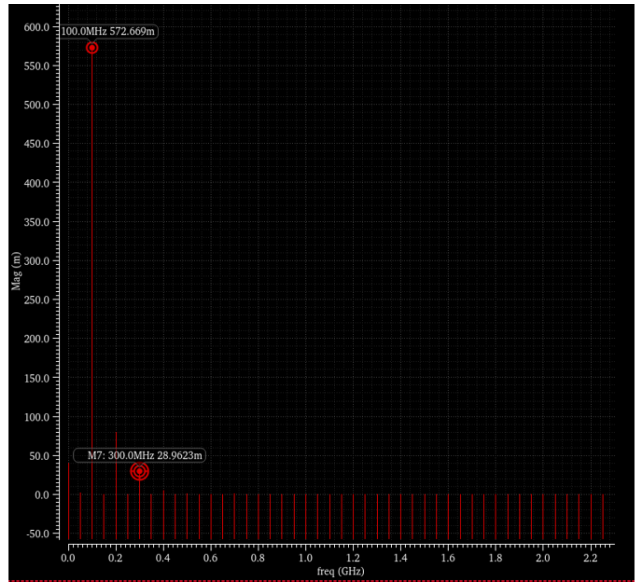

Figure 3: DFT spectrum analysis for the Active Load configuration

3 Final Optimized Architecture

Confronted by the biasing instability and high distortion of the active load framework, the design reverted to a single-transistor NMOS Common-Source configuration, focusing on deep passive source degeneration to meet the linearity goal.

3.1 Circuit Topology and Component Sizing

The final optimized schematic features a structurally robust, highly stable configuration:

- **Gate Biasing Layout:** Passive resistors $R_{\text{bias1}} = 10$ kΩ and $R_{\text{bias2}} = 5.5$ kΩ establish a reliable quiescent gate voltage ($V_G \approx 0.638$ V), locking the device securely into the saturation region.
- **AC Coupling Capacitors:** C_{in} and C_{out} are sized at the maximum allowed boundary of 100 pF. At 100 MHz, their reactive impedance is small ($|Z_C| \approx 15.91\ \Omega$), ensuring transparent AC signal transmission.
- **Passive Loading & Degeneration:** The passive drain resistor (R_D) is set to 270 Ω. Linearity correction is handled by a 40 Ω source degeneration resistor (R_S).

3.2 Transistor Dimensioning

To compensate for the gain reduction caused by the larger degeneration resistor, a wide-channel transistor geometry was selected:

- **Width** **(**W**):** 550 μm
- **Length** **(**L**):** 180 nm
- **Multiplier** **(**m**):** 1

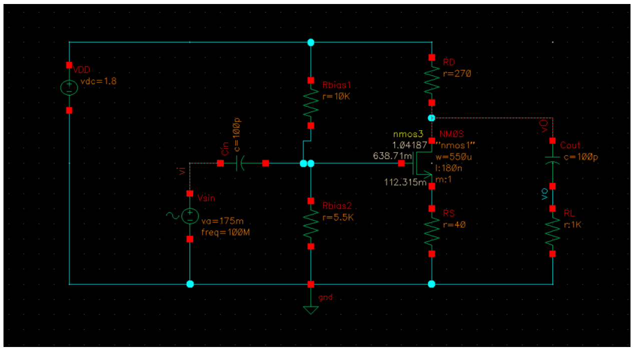

Figure 4: Schematic diagram of the final optimized degenerated NMOS Common-Source Amplifier design.

4 Simulation Results and Analysis

The final single-transistor architecture was evaluated with a 100 MHz input tone ($V_a = 175$ mV, providing a 350 mV peak-to-peak input envelope).

4.1 Transient Evaluation

The output AC voltage waveform (v_O) oscillates symmetrically on a stable DC quiescent baseline, moving between a minimum value of ≈ 0.55 V and a maximum value of ≈ 1.47 V. This generates an effective peak-to-peak output swing of:

$$V_{o,\text{pk-pk}} \approx 1.47 \text{ V} - 0.55 \text{ V} = 0.92 \text{ V} \tag{3}$$

This swing matches the target specifications across the external 1 kΩ AC-coupled load resistor.

4.2 Spectral Harmonics, Gain, and Distortion

Applying a Discrete Fourier Transform (DFT) to the steady-state transient output yields the following harmonic magnitudes:

- **Fundamental Tone ($HD1$) at 100 MHz:** 465.94 mV
- **Third Harmonic ($HD3$) at 300 MHz:** 5.486 mV
- **Voltage Gain (A_v):** Given an input drive peak amplitude $V_a = 175$ mV and a fundamental frequency output peak component of 465.94 mV, the linear voltage gain yields:

$$A_{v,\text{final}} = \frac{465.9447 \text{ mV}}{175 \text{ mV}} \approx 2.66 \text{ V/V } (8.50 \text{ dB}) \tag{4}$$

The final suppression performance of the optimized circuit is computed as:

$$\left(\frac{HD1}{HD3}\right)_{\text{final}} = 20\log_{10}\left(\frac{465.9447 \text{ mV}}{5.486496 \text{ mV}}\right) \approx 38.58 \text{ dB} \tag{5}$$

By choosing a 40 Ω source degeneration resistor instead of the 10 Ω variant used in the active-load phase, the distortion profile improved from 25.92 dB to ≈ 38.6 dB, approaching the target limit while maintaining a large output voltage envelope.

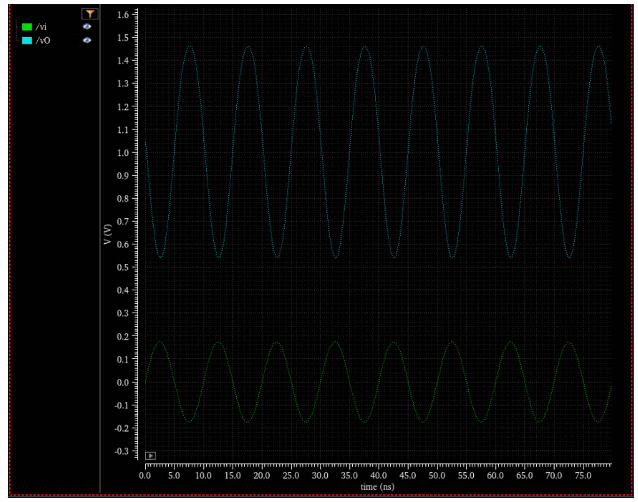

Figure 5: Transient waveform profile for the final optimized degenerated Common-Source configuration.

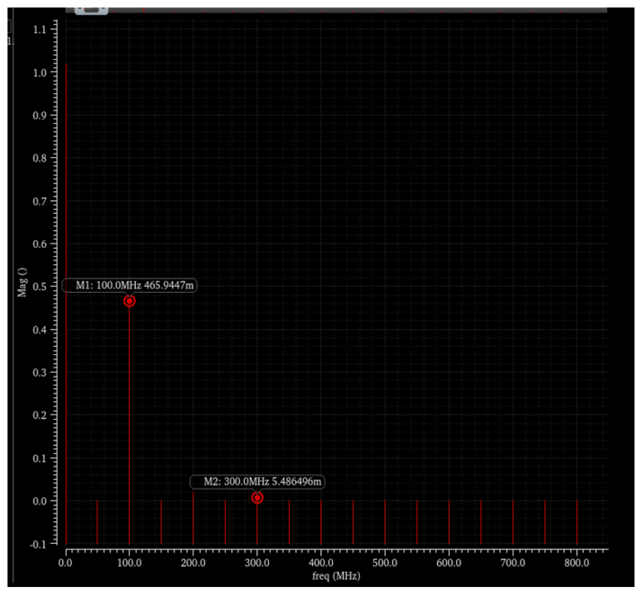

Figure 6: DFT spectrum analysis for the final optimized degenerated Common-Source configuration.

5 Comparative Analysis

A direct comparison highlights the strict trade-offs observed during the architectural exploration phases. The Phase 2 CMOS active-load pair was initially designed to yield large signal headroom, producing a fundamental output amplitude ($HD1$) of 572.67 mV. However, due to severe biasing shifts under high-swing regimes, it suffered from a high third-harmonic distortion component ($HD3 = 28.96$ mV), resulting in a poor linearity metric of 25.92 dB and a degraded fundamental voltage gain of only 1.27 V/V (2.09 dB).

In contrast, reverting to the passive degenerated NMOS structure provided robust negative feedback loop stability. Although the final design yields a slightly lower fundamental amplitude of 465.94 mV, it vastly reduces third-harmonic generation down to 5.49 mV. This optimization shifts the $HD1/HD3$ ratio up to 38.58 dB, successfully maximizing the fundamental large-signal voltage gain to 2.66 V/V (8.50 dB) while offering full operational tracking symmetry across the external network.

6 Conclusion

A high-linearity 100 MHz common-source amplifier was designed and verified using a 180nm CMOS process node. The design journey demonstrated that while active current-source loads provide high gain, they present biasing challenges and poor harmonic performance under high-swing conditions. Returning to a single NMOS configuration with an optimized $W/L = 550\ \mu\text{m}/180$ nm and a $40\ \Omega$ source degeneration resistor resolved these issues, delivering a reliable 0.92 V peak-to-peak output swing and an improved $HD1/HD3$ ratio of ≈ 38.6 dB.